

SYSTEMS TOOL KIT (STK) SOFTWARE RENEWAL 2025

Requirements Document

For

704th Test Group/746th Test Squadron

10 March 2025

1.0 PURPOSE

The 704th Test Group (704 TG) mission is to be the premier provider of unique test and evaluation capabilities relevant to tomorrow's warfighter through world-class people improving technical excellence, cost-effectiveness, and agility. The 746th Test Squadron (746 TS) is assigned to the 704 TG.

The 746 TS analysis flight uses the physics-based software package Systems Tool Kit (STK) from Ansys for the engineers and scientists to perform complex analysis of land, sea, air, and space assets, and share results in one integrated solution. STK is designed to determine the time-dynamic position and attitude of assets, determining dynamic spatial relationships among all of the objects under consideration including the quality of those relationships or accesses given a number of complex, simultaneous constraining conditions.

STK provides 746 TS engineers the capability for conjunction analysis, maneuver modeling, orbit trajectory modeling, orbit determination, system performance analysis, aircraft and UAV modeling, communication link analysis, sensor modeling, antenna modeling, terrain modeling, mission modeling and atmospheric effects. This is a unique software modeling environment that provides the 746 TS with verified results in the test planning, execution, post flight analysis, safety of flight assessments, radio frequency (RF) interference data in a time dynamic data analysis engine that no other vendor can provide.

2.0 SCOPE

STK Software renewals for lab and analysis workstations for one (1) year

Part Number	Title/Name	Quantity
ASTKPre-A-TECS	Ansys STK Premium (Air) – Incorporate aerodynamics and performance profiles for highly accurate flight profiles (TECS)	1

Services and support/maintenance included.

3.0 MILESTONES

The contractor shall deliver the STK software renewal no later than 10 days ARO.

4.0 DELIVERABLES

The contractor shall deliver all hardware and documentation to 746 TS/CZF, Attn: Carlos A. Ruiz, at 1644 Vandergrift Road, building 1265, Holloman Air Force Base, New Mexico 88330. Commercial Phone: (575) 572-0212